14 Mark for Review

One of the factors of $2x^3 + 44x^2 + 234x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b?

Answer Preview:

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Which expression is a factor of $y^2(x-9)-36(x-9)^3$?

A. $y(x-9)$

B. $(x-9)(x-6)$

C. $y+x-9$

D. $y+6x-54$

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$$r(x) = 13(x - 2)$$

$$s(x) = x^3 + nx^2 + 2nx + 8$$

For the given functions r and s, n is a constant. If $r(x) \cdot s(x) = 13(x^4 - 16)$, what is the value of n?

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The function g is defined by $g(x) = (x + 13)(t - x)$, where t is a constant. In the *xy*-plane, the graph of $y = g(x)$ passes through the point $(24, 0)$. What is the value of $g(0)$?

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$$f(x) = 2x + 3$$

$$g(x) = 7x - 2$$

$$h(x) = 5x + 6$$

The functions f, g, and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a, b, and c are constants, what is the value of b?

A. -5

B. 12

C. 20

D. 22

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$$(4x + 10)(2x + 1) - (2x - 14)$$

For all values of x, the given expression is equivalent to which of the following?

I. $8x^2 + 22x + 24$
II. $8x^2 + 24x + 10 - 2x - 14$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

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3 Mark for Review ABC

Which expression is a factor of $x^4 + 14ax^2 + 49a^2 - 64$, where a is a positive constant?

(A) $x^2 + 7a + 8$

(B) $x^2 - 7a - 8$

(C) $x^2 + 14a$

(D) $x^2 - 8a$

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7 Mark for Review

Which expression is a factor of $y^2(x-3)-25(x-3)^3$?

(A) $y(x-3)$

(B) $(x-5)(x-3)$

(C) $y+x-3$

(D) $y+5x-15$

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$$f(x) = 11x^{10} + 3x^8$$

$$g(x) = -17x^7 + 9x^5$$

The polynomial $p(x)$ is defined as the product of the given polynomials, $f(x)$ and $g(x)$. What is the coefficient of x^{15} in $p(x)$?

(A) 15

(B) 48

(C) 99

(D) 150

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15 Mark for Review ABC

Which of the following is a factor of the expression $2x^2 + (18r + 5)x + 45r$, where r is a nonzero constant?

I. $x + 9r$

II. $2x + 5r$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

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15 Mark for Review

One of the factors of $8x^4 + 50x^2 + 63$ is $ax^2 + b$, where a and b are positive constants. Which of the following is a possible value of ab?

(A) 7

(B) 14

(C) 18

(D) 36

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9 Mark for Review

Which expression is a factor of $49p^{19} - 121p^{17}$?

(A) $7p + 11$

(B) $49p^2 + 121$

(C) $11p - 7$

(D) $-72p^{40}$

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$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a, b, and c are integer constants. The zeros of the function are -2, -7, and 6. What is the value of a?

(A) -84

(B) -3

(C) 3

(D) 84

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